Bernhard Haslhofer

Co-Founder & Head of Research, Iknaio Cryptoasset Analytics GmbH
Faculty & Research Group Leader, Complexity Science Hub Vienna

bernhard.haslhofer@iknaio.com | iknaio.com | bernhardhaslhofer.info

Computer scientist with 10+ years of research on cryptoasset analytics and financial crime detection, and co-founder of Iknaio, which turns this research into investigation products for regulators, law enforcement, and financial institutions. Creator of the open-source GraphSense platform that underpins Iknaio's technology, PI and coordinator of national and EU projects on cryptoasset forensics and AI for investigations, and author of 80+ peer-reviewed papers (h-index 37).

Key Achievements

- **Research-to-market transfer** – Co-founded Iknaio (2021) as a spin-off of publicly funded research; awarded the **Houska Prize 2024** (2nd place), Austria's premier award for innovative IT solutions, for this transfer.
- **GraphSense creator** – Open-source cryptoasset analytics platform with 1,000+ users in research, law enforcement, and compliance; now hosted and developed further by Iknaio.
- **EUR 3M+ in competitive funding** – PI or coordinator of FFG KIRAS, FFG Bridge, and EU Horizon projects on blockchain forensics, DeFi crime, and LLMs for investigations.
- **Trusted by public authorities** – Technical Advisor to the Bank for International Settlements; Host & Academic Chair of the INTERPOL New Technologies Forum (2025); close collaboration with Austrian ministries (BMI, BMF, BMJ), FMA, and OeNB.
- **Research leadership** – Leads the Digital Currency Ecosystems group (6 FTEs) at the Complexity Science Hub; previously built and led the Data Science research field at AIT.
- **Open standards** – Founding editor of the Open Blockchain Intelligence Standards (OBIS), vendor-neutral specifications for attribution and data exchange between investigators, supervisors, and tool providers.

Professional Experience

2021 – Present	**Co-Founder & Head of Research**, *Iknaio Cryptoasset Analytics GmbH*. Leads research and development of cryptoasset analytics products and services; transfers research results into production.
2021 – Present	**Faculty & Research Group Leader**, *Complexity Science Hub, Vienna*. Computational methods for analyzing digital currency ecosystems, including DeFi.
2022 – Present	**Technical Advisor**, *Bank for International Settlements – Innovation Hub Eurosystem Centre*.
2013 – 2022	**Scientist → Lead of Research Field Data Science**, *AIT Austrian Institute of Technology*. Established the cryptoasset analytics research line and the Data Science group.
2011 – 2014	**Postdoctoral Associate (EU Marie Curie Fellow)**, *Cornell University, USA*.
2004 – 2011	**Researcher / Assistant Professor**, *University of Vienna; Austrian Research Centers*.

Education

2025	**Habilitation in Computer Science**, *TU Wien*.
2008	**PhD in Computer Science** (with distinction), *University of Vienna*.
2003 / 2005	**Diploma in Economics & Computer Science; MSc in Information & Knowledge Management**, *TU Wien*.

Selected Grants

2026 – 2028	**DIFPILOT** (FFG Bridge), *Coordinator*. LLM-assisted workflows for digital forensic investigators.
2025 – 2027	**LLEA** (FFG KIRAS), *Coordinator*. LLM-based data analysis methods for law enforcement.
2024 – 2027	**ENSEMBLE** (Horizon Europe CL3), *Partner*. AI-based technology against cross-border cybercrime.
2023 – 2025	**DeFiTrace** (FFG KIRAS), *Coordinator, PI*. Detecting illegal transactions in cross-ledger DeFi systems.
2023 – 2025	**AMALFI** (FFG Bridge), *Coordinator, PI*. Graph learning on large multilayer cryptoasset transaction networks.
2020 – 2022	**KRYPTOMONITOR** (FFG KIRAS), *Coordinator, PI*. Analysis of smart contracts and off-chain transactions.
2017 – 2020	**TITANIUM** (EU Horizon 2020), *Senior Scientist*. Data-driven tools for investigating crimes involving virtual currencies.
2015 – 2017	**GraphSense** (FFG ICT of the Future), *Coordinator, PI*. Origin of the GraphSense platform.

Selected Publications

1. Avice, R., **Haslhofer**, **B.**, Li, Z., & Zhou, J. (2026). Linking cryptoasset attribution tags to knowledge graph entities: An LLM-based approach. In C. Garman & P. Moreno-Sanchez (Eds.), *Financial cryptography and data security* (pp. 366–382). Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-07035-7_22

2. Carpentier-Desjardins, C., Paquet-Clouston, M., Kitzler, S., & **Haslhofer**, **B.** (2025). Mapping the DeFi crime landscape: An evidence-based picture. *Journal of Cybersecurity*, *11*(1), tyae029. https://doi.org/10.1093/cybsec/tyae029

3. Kitzler, S., Victor, F., Saggese, P., & **Haslhofer**, **B.** (2023). Disentangling decentralized finance (DeFi) compositions. *ACM Trans. Web*, *17*(2). https://doi.org/10.1145/3532857

4. Kappos, G., Yousaf, H., Stütz, R., Rollet, S., **Haslhofer**, **B.**, & Meiklejohn, S. (2022). How to peel a million: Validating and expanding Bitcoin clusters. In K. R. B. Butler & K. Thomas (Eds.), *31st USENIX Security Symposium, Boston, MA, USA, August 10-12, 2022* (pp. 2207–2223). USENIX Association

5. Romiti, M., Victor, F., Moreno-Sanchez, P., Nordholt, P. S., **Haslhofer**, **B.**, & Maffei, M. (2021). Cross-layer deanonymization methods in the Lightning protocol. In N. Borisov & C. Díaz (Eds.), *FC 2021 - 25th international conference on financial cryptography and data security* (pp. 187–204, Vol. 12674). Springer. https://doi.org/10.1007/978-3-662-64322-8_9

6. Fröwis, M., Gottschalk, T., **Haslhofer**, **B.**, Rückert, C., & Pesch, P. (2020). Safeguarding the evidential value of forensic cryptocurrency investigations. *Forensic Science International: Digital Investigation*. https://doi.org/10.1016/j.fsidi.2019.200902